

Why Sampling?

Reading Material: Chapter 12 of KF book

Very often we need to sample instances from a joint distribution

$P(\mathbf{y}_1, \dots, \mathbf{y}_n | \mathbf{x})$ or $P(\mathbf{x} = x_1, \dots, x_n)$ or

$P(x_1, \dots, x_r | x_{r+1} = E_{r+1}, \dots, x_n = E_n)$. Here are some scenarios:

- During training we need to solve an intractable inference
- We need to show a diverse set of outputs to a user instead of just the most likely value. Example: in translation.
- We need to calculate expected value of some arbitrary function $f(\mathbf{x})$ under distribution $P(\mathbf{x})$. What is the expected number of times that adjacent positions have the same label for a given $\mathbf{x}$?
 - Modify the objective value of a trained model to incorporate a user-specified reward function.

Motivation: Inference Deep Language Models

- Generate sample sentences
- Generate questions
- Expected distribution of first word for sentences ending with '?'.


Sampling to approximate expected value of a function under $P(\mathbf{x})$

Let $\mathcal{X}$ denote the space of $\mathbf{x}$, example space of sentences

Given $f(\mathbf{x}) \mapsto R$ denote a function. Our goal is to calculate the expected value of this function under $P(\mathbf{x})$. Example, fraction of sentences that are longer than 7 words.

$$f(\mathbf{x}) = \begin{cases} 1 & \text{if } |\mathbf{x}| \geq 7 \\ 0 & \text{otherwise} \end{cases}$$

$$\mu_f = E_{p(\mathbf{x})}(f(\mathbf{x})) = \sum_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) P(\mathbf{x})$$

The above is intractable when $\mathcal{X}$ is large. Approximate with samples. Let $S = \mathbf{x}^1, \dots, \mathbf{x}^M$ denote samples from $P(\mathbf{x})$

$$\mu_f \approx \hat{\mu}_f(S) = \frac{1}{M} \sum_{i=1}^M f(\mathbf{x}^i)$$

Basics: What are desired samples

Let $p(x)$ be a distribution. If we draw a set S of M samples $x^1, \dots, x^M$ from the distribution, what is the characteristic of a good sampling algorithm?

$$x \sim \mathcal{N}(x; 10; 1) \rightarrow p(x) \quad M=5$$

$$\begin{array}{rcl}
 S^1: & 10.1 & -100, \quad 20, \quad -20, \quad 0 \\
 S^2: & 9.0 & 10.9 \quad 12 \quad 8 \quad 3 \\
 S^3: & -20 & 20 \quad -10 \quad 20.1 \quad 11.
 \end{array}$$

✓

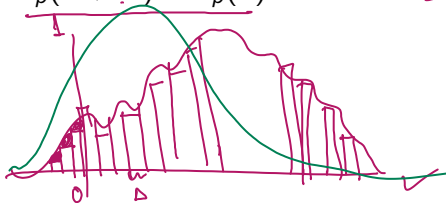
$$\mu(S^2) =$$

Consistent samples

Let $F_p(x)$ denote the cumulative distribution of $p(x)$

Continuous distributions: As $M \rightarrow \infty$, the fraction of times in the sample S that we encounter a sample in an interval $[x, x + \Delta)$ would be proportional to the true probability of that interval in $p(x)$ that is,

$$F_p(x + \Delta) - F_p(x)$$



$$S \equiv \{x^1, x^2, \dots, x^M\}$$

$$n(x \in \Delta, x, S) = \frac{\sum_{i=1}^M \delta(x^i \in [x, x + \Delta])}{M}$$

Consistent samples: discrete

$$x \in \{1, 2, \dots, k\}$$

$$p(x) = \text{cat}(x; \pi_1, \pi_2, \dots, \pi_k) \quad \text{g.f.}$$

$$S \equiv \{x^1, x^2, \dots, x^M\}$$

$$n(j, S) =$$

$$\frac{\sum_{i=1}^M \delta(x^i = j)}{M}$$

$$\left(\sum_{j=1}^k |n(j, S) - \pi_j| \rightarrow 0 \right)$$

then S is
consistent.

Consistent estimates

The estimated expected value from a sample S is consistent if

$$\lim_{M \rightarrow \infty} \hat{\mu}_f(S) = \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{i=1}^M f(\mathbf{x}^i) \rightarrow \underline{\mu_f}$$

Unbiased samples

A sampling algorithm A is unbiased if the probability of it selecting an element x in S is proportional to its density $p(x)$ or probability $P(x)$ irrespective of the size of the sample M .

Checking for bias. For any finite M , get a large number N of samples $S_1, \dots, S_N$ each of size M . The sampling algorithm A is unbiased if

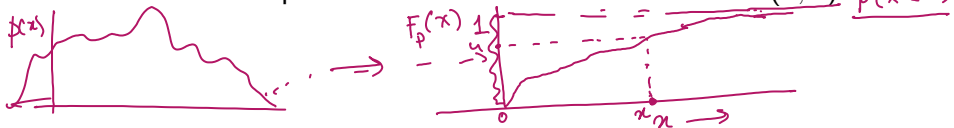
$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^N \hat{\mu}_f(S_j) \rightarrow \mu_f \quad (1)$$

Basics: Sampling scalar distributions



Let $p(x)$ be a distribution. An unbiased consistent algorithm for drawing M samples $x^1, \dots, x^M$ from the distribution is:

Assume we can sample a u from a uniform distribution $U(0, 1)$



Let $F_p(x)$ be cumulative distribution of $p(x)$.

For $i = 1 \dots M$

- 1 Sample $u_i \sim U(0, 1)$
- 2 Find $x^i = F^{-1}(u)$

Basics: Sampling from multinomial distributions

x is discrete, $x \in \{1, \dots, m\}$

$$p(x) \sim \text{Mult}(p_1, \dots, p_m)$$

$$F_p(x) = \sum_{i=1}^x \pi_i$$

for $i = 1$ to M

$u_i \sim U(0, 1)$

j s.t. $F_p(j-1) \leq u_i < F_p(j)$

$x^i = j$

}

Make sure corner cases are right.

$$p(x) = \text{Cat}(x; [0.3, 0.5, 0.2])$$

$$F_p(x=1) = 0.3$$

$$F_p(x=2) = \pi_1 + \pi_2$$

$$= 0.8$$

$$F_p(x=3) = 1$$

How to sample multivariate distributions?

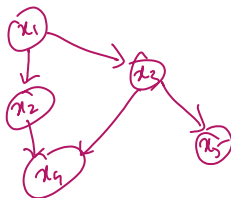
Option 1: Factorize the distribution as a Bayesian network and perform forward sampling.

$$P(x) \equiv LLM$$

$$x \equiv x_1, x_2, \dots, x_n$$

$$P(x) = \prod_{j=1}^n P(x_j | x_1, x_2, \dots, x_{j-1})$$

Example 2



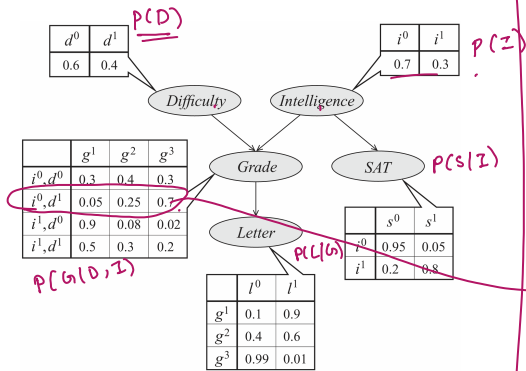
Example 1



Forward Sampling Algorithm

- ① For $i = 1$ to M
 - ① Let $\underline{j_1, \dots, j_n}$ denote a topological sort order of $\underline{n}$ variables as per G
 $\{^1, \{^2 \dots \dots \xi^M$
 - ② For $\underline{k = 1}$ to $\underline{n}$
 - ① $\underline{\xi_{j_k}^i} \sim \underline{P(x_{j_k} | Pa(j_k)) = \underline{\xi_{Pa(j_k)}^i)}$
 - ③ Return $\underline{\xi^1 \dots \xi^M}$

Example: Forward sampling



$\delta_1, \delta_2, \dots, \delta_5 = \{D, I, G, S, L\}$

$i = 1$
 $\xi^i = [\phi, \dots, \phi]$
 $k = 1$
 $\xi_1^i \sim P(D)$ — $u \sim \{0, 1\}$
 $= 0.7$
 $\xi_1^i = d_1$

$k = 2$
 $\xi_2^i \sim P(I)$ — $u \sim \{0, 1\}$
 $= 0.3$
 $\xi_2^i = i^0$

$k = 3$
 $\xi_3^i \sim P(G|D=d_1, I=i^0)$ — $u \sim \{0, 1\}$
 $= 0.6$
 $\xi_3^i = g_3$

Importance Sampling

- What fraction of sentences end with '!!!'? In forward sampling most of the sampled sentences would be wasted since they would not end with '!!!'.
- Complete missing attribute in a VAE network for object generation. Forward sampling would not match given values most of the time.
- In general: importance sampling is useful when it is hard to sample from $P(\mathbf{x})$ or to lower the error in computation of expected value of a function.

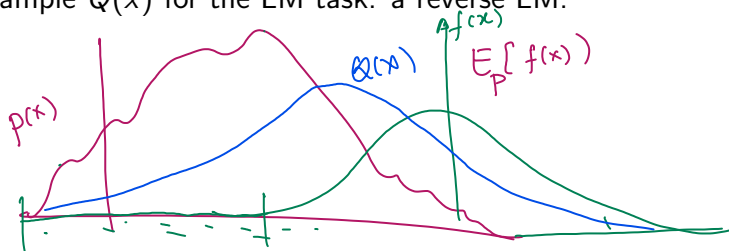
$$E[f(\mathbf{x})] = \sum_{\mathbf{x}} P(\mathbf{x}) \underline{f(\mathbf{x})}$$

where $f(\mathbf{x})$ is zero for many $\mathbf{x}$. Example, rare combinations.
Importance sampling — sample from the *important* regions.

Estimation with importance sampling.

Proposal distribution: $Q(x)$ from which it is easy to generate samples. Designing a good proposal distribution is an 'art' and problem-dependent.

Example $Q(x)$ for the LM task: a reverse LM.



Estimation with importance sampling.

- Get M samples S_Q from $Q(\mathbf{x})$: $\mathbf{x}^1$, ..., $\mathbf{x}^M$
-

$$E_p[f(\mathbf{x})] \approx \mu_f(\hat{S}_Q) = \frac{1}{M} \sum_{i=1}^M f(\mathbf{x}^i) \frac{P(\mathbf{x}^i)}{Q(\mathbf{x}^i)}$$

- Example:

$P(\mathbf{x}) \equiv$ sentence generator from left to write
 $Q(\mathbf{x}) \equiv$ " " " right to left.
 M sentences using $Q(\mathbf{x}) \rightarrow S_Q$
 $f(\mathbf{x}) \equiv \begin{cases} 1 & \text{if } \mathbf{x} \text{ ends with "!!!"} \\ 0 & \text{otherwise.} \end{cases}$

$$\hat{\mu}_f(S_Q)$$

Importance sampling is consistent.

$$S_Q = \{x^1, x^2, \dots, x^M\}$$

$$M \rightarrow \infty$$

$$= \mu_f = \sum_{x \in \mathcal{X}} f(x) P(x)$$

$$\lim_{M \rightarrow \infty} \hat{\mu}_f[S_Q] = \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{i=1}^M f(x^i) \frac{P(x^i)}{Q(x^i)}$$

$$\frac{1}{M} \sum_{i=1}^M g(x_i) = \sum_{x \sim Q} Q(x) g(x)$$

① because S_Q is consistent.

$$= \sum_{x \sim Q} \left[f(x) \frac{P(x)}{Q(x)} \right] Q(x)$$

$$= \sum_{x \sim Q} f(x) P(x) \stackrel{g(x)}{=} \mu_f$$

Normalized importance sampling.

Sometime $P(x)$ is available only in un-normalized form

Example 1: $P(x)$ is an undirected graphical model where z is difficult to compute. $\frac{\prod_{c \in \text{clique}} \psi_c(x_c)}{z}$

Example 2: $P(x)$ is expressed in terms of a energy based model $P(x) = \frac{e^{-E_\theta(x)}}{z}$

$$P(x) = \frac{\tilde{P}(x)}{z}$$

We have available only $\tilde{P}(x)$

$$\hat{u}_f(S_a) = \frac{1}{M} \sum_{i=1}^M f(x^i) \frac{P(x^i)}{Q(x^i)} = \frac{1}{M} \sum_{i=1}^M f(x^i) \frac{\tilde{P}(x^i)}{z Q(x^i)} = \frac{1}{M \cdot z} \sum_{i=1}^M f(x^i) \tilde{P}(x^i)$$

$$z = \sum_{x \in \mathcal{X}} \tilde{P}(x) = \sum_{x \in \mathcal{X}} \tilde{P}(x) \frac{Q(x)}{Q(x)} = \sum_{x \in \mathcal{X}} \left[\underbrace{\frac{\tilde{P}(x)}{Q(x)}}_{q(x)} Q(x) \right] \stackrel{E[g]}{=} \frac{1}{M} \sum_{i=1}^M \frac{\tilde{P}(x^i)}{Q(x^i)}$$

Estimation with normalized importance sampling.

- Get M samples S_Q from $Q(\mathbf{x})$: $\mathbf{x}^1, \dots, \mathbf{x}^M$
- Compute weight of each sample as $\tilde{w}_i = \frac{\tilde{P}(\mathbf{x}^i)}{Q(\mathbf{x}^i)}$
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$$\mu_f(\hat{S}_Q) = \frac{\sum_{i=1}^M f(\mathbf{x}^i) \tilde{w}_i}{\sum_{i=1}^M \tilde{w}_i}$$

Application of Normalized importance sampling in Reward-based sampling from forward sampling models

Suppose we have a distribution $P(\mathbf{x}) = P(x_1, \dots, x_n)$ from which we can easily sample using the forward sampling method.

Now your goal is to impose a cost on $\mathbf{x}$ and sample from

$$P_C(\mathbf{x}) \propto P(\mathbf{x}) \exp(-C(\mathbf{x}))$$

We can use importance sampling as follows: